A Heads Up Technologies Company

Troubleshooting CMS and IFE Networks

SEE YOU AT AEA DALLAS 26

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Syllabus

Audio

- Speaker Mounting Locations
- Installation Techniques
- System Checkout
- Audio System Tuning

Cabin Management Systems

- Creating a thorough test plan
- Isolating the problem
- Using diagnostic tools

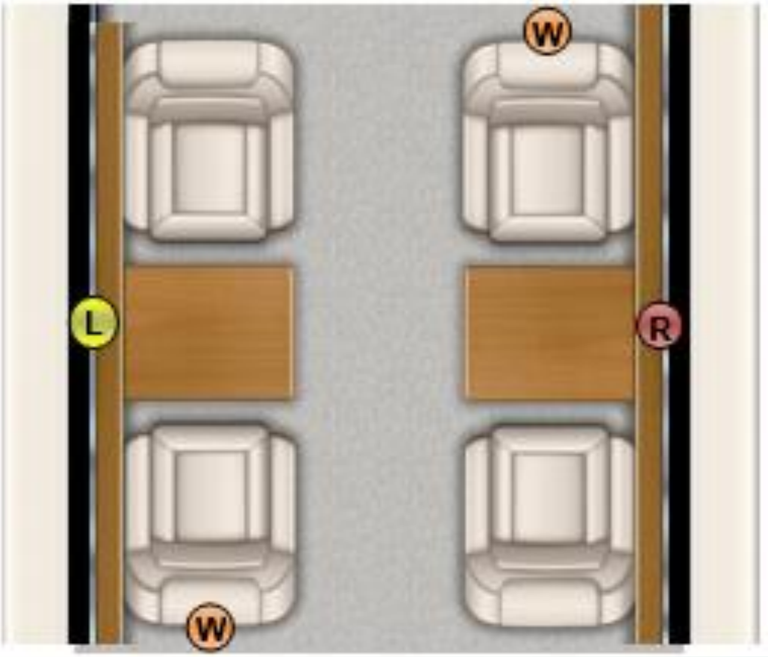
Networking

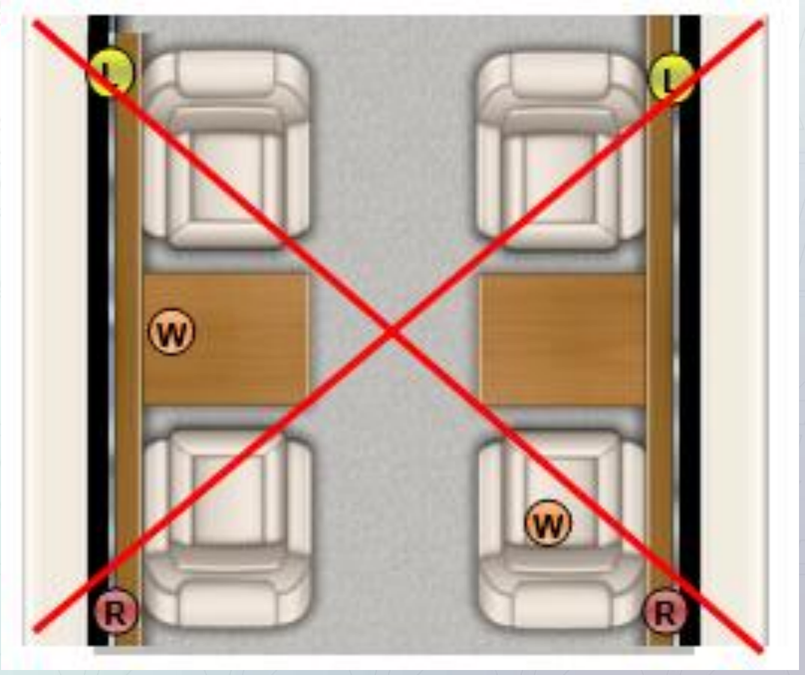
- Ethernet
- Addressing
- Capturing Network Traffic
- Wireshark

Speaker Mounting Location Considerations

Stock locations are okay to retain if you are trying to limit work scope, but usually not ideal.

1. Main left and right speakers should be in front
2. Surround speakers behind
3. Subwoofers in the corners



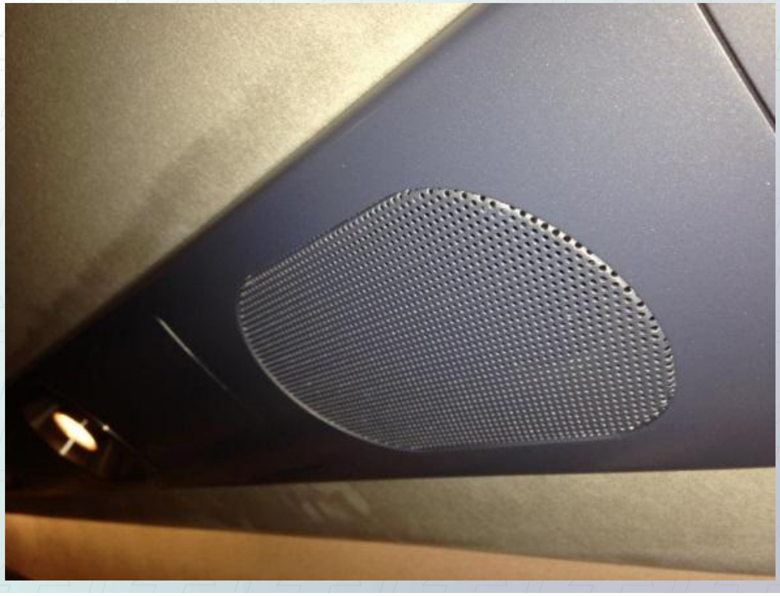


Speaker Installation Techniques

Grill Material Selection:

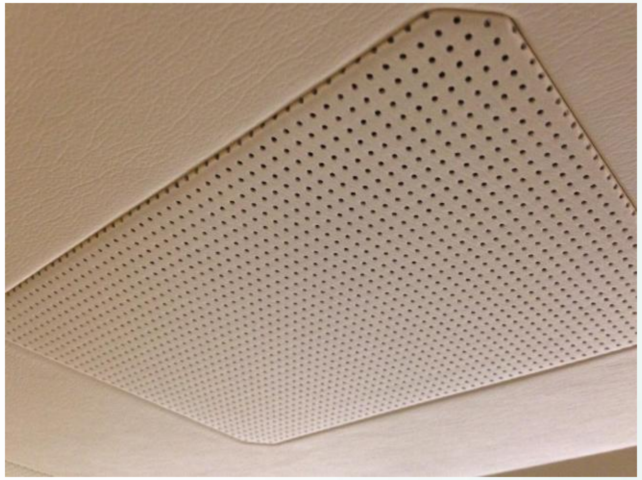
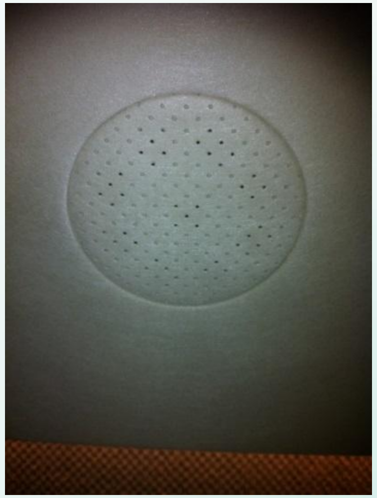
- Plastic or metal is ideal.
- If fabric, it should be acoustically transparent.
- UltraSuede or leather are not good speaker grill materials, even with holes.

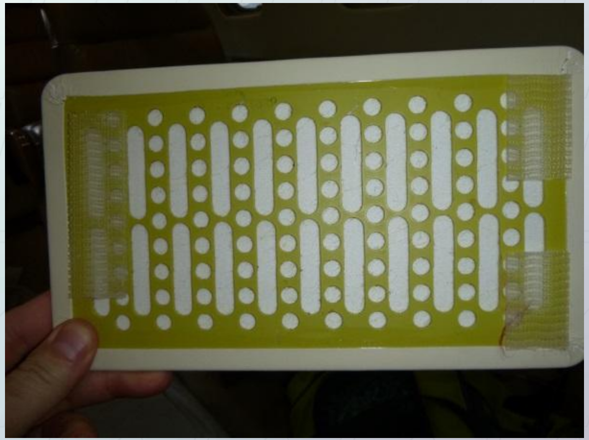
High quality metal grills



Speaker Installation Techniques

Useful Tip: Not sure if a material is acoustically transparent?
Try to blow air through the material. If air does not pass through with ease, then this material is NOT acoustically transparent and would make a poor speaker grill

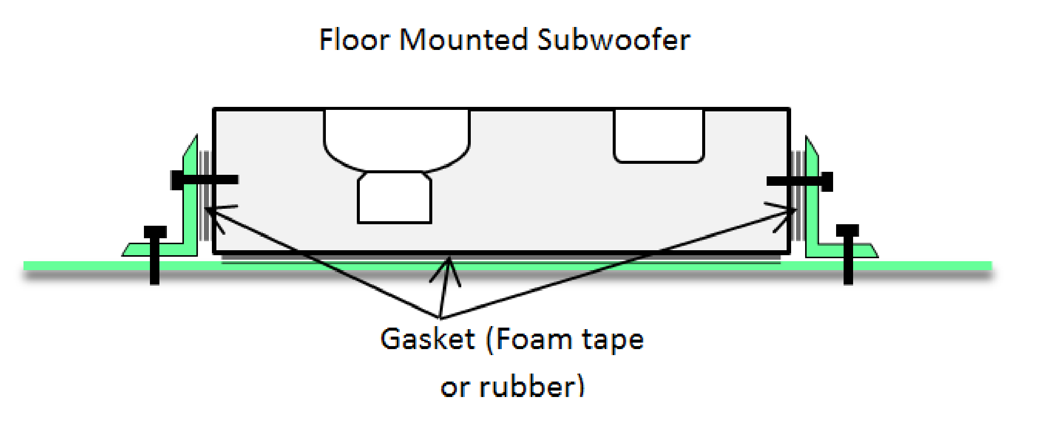


Speaker Installation Techniques

When mounting mid-high frequency loudspeakers, be sure to use gasket material around the edge of the speaker. This will eliminate any potential for buzzing.

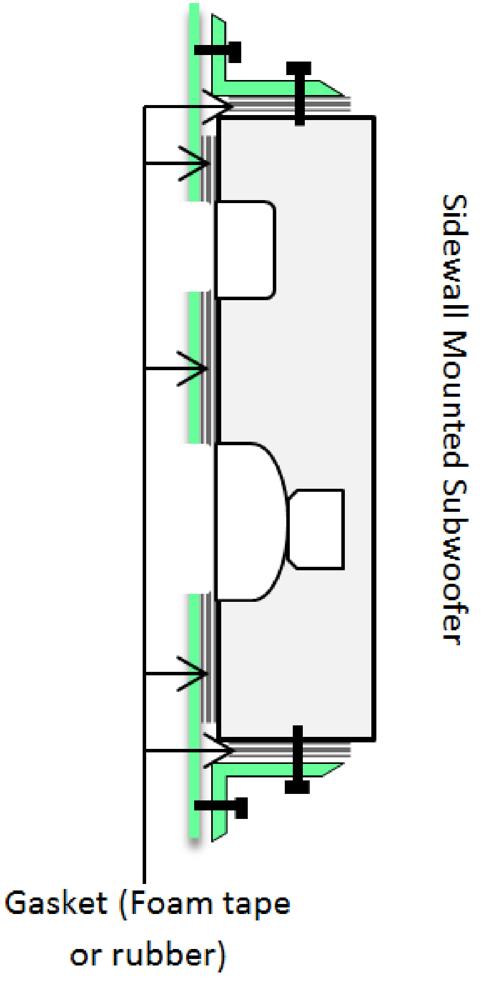
Consider the use of gasketing on cabinet doors and drawers adjacent to subwoofers.

Subwoofer Installation Techniques



Air Path: Open Grilles and Ports

Vibration: Include gasket material in between any surfaces the enclosure may touch.



Functional System Checkout

Play each speaker individually – Confirm expected location

Automated with ALTO's Forte Software, or in a traditional system you could unplug speakers and plug in one at a time to confirm proper channel routing.

Functional System Checkout (cont.)

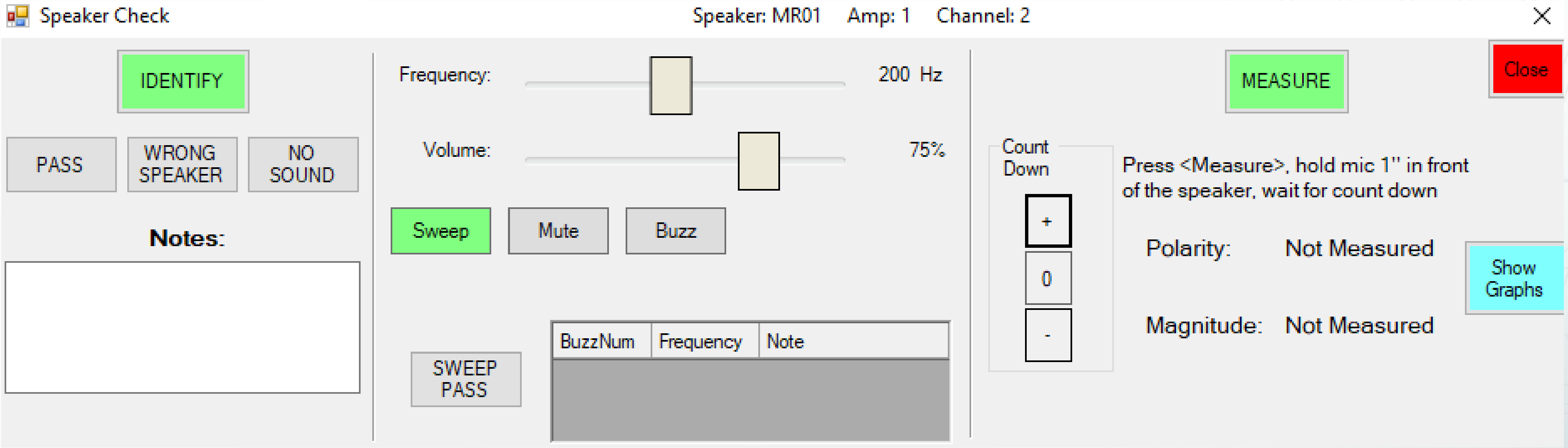
Channel Mapping Testing

- Play left/right test track for stereo systems. Play 5.1 (or higher) test track for surround sound systems.
- Test left/right/5.1 for every audio source available. Bluetooth, HDMI, Analog etc.

PA Testing

- Test all microphones, sidetone, SELCAL, briefer and call functions
- Determine if there are any special routing or customer requirements (like Call chime limited to certain areas)
- Set levels considering ambient noise in flight

Functional System Checkout (cont.)

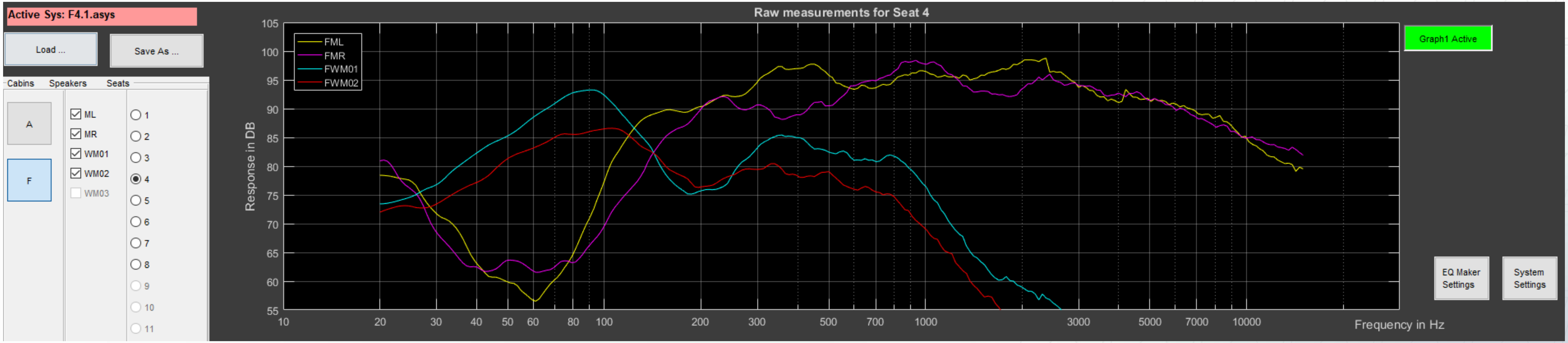


Rub and Buzz Testing

- Use a tone generator app to play a single frequency at a medium to high level.
- Slowly scroll through the entire frequency spectrum listening for buzzing. If you hear buzzing, address it by de-coupling hard materials.

Shown above is tone generator in ALTO Forte software, but free tone generator apps are available in Apple App Store or Google Play Store for mobile devices.

Audio Tuning



Custom ALTO tuning performed by an ALTO's system engineer utilizing custom EQDesigner software, or manual tuning using RTA and equalizer or another DSP unit.

Set audio levels for all sources so switching between devices sounds uniform.

Cabin Management System Architecture

Inputs:

- Switches
- App
- Media

Network

Controllers:

- Relays
- Potentiometer
- DAC

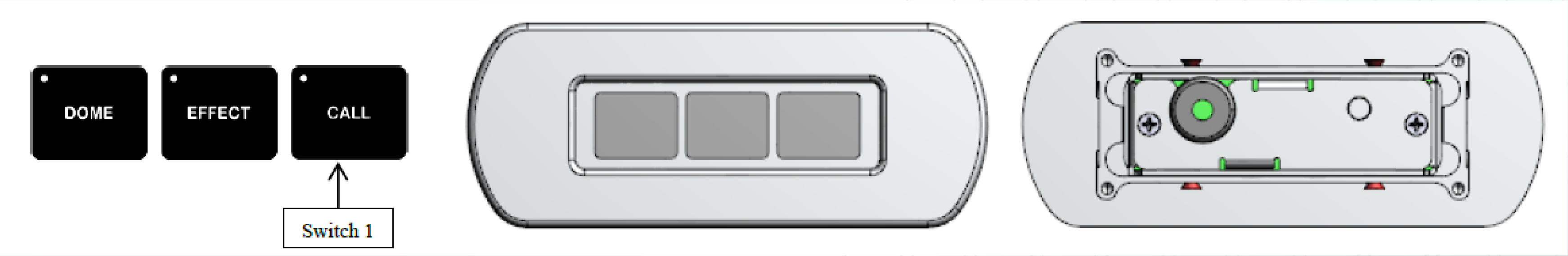
Network

Loads:

- Lights
- Temp
- In Flight Entertainment

Cabin Management Documentation

Configuration Document: Switches



Interface Definition: Controllers

Conn	Pin	Function	Circuit Assignment	Notes	Software Function	Status Output	Target Relay
J2	19	INPUT A1	READ LIGHT SWITCH		TOG	SO1	RELAY 1
	37	INPUT A2	TABLE LIGHTS SWITCH		TOG	SO2	RELAY 2
	18	INPUT A3	FLUSH SWITCH		MOM	SO3	RELAY 3
	24	STATUS OUTPUT 1	READ LIGHT SWITCH STATUS LED				
	5	STATUS OUTPUT 2	TABLE LIGHT SWITCH STATUS LED				
J3	17	OUTPUT 1 NC-A	READ LIGHT RELAY				
	18	OUTPUT 1 NO-A		GND to READ LIGHT (-)			
	36	OUTPUT 1 NC-B					
	37	OUTPUT 1 NO-B		Connect to GND			

... Layout | RC1 | RC2 | RC3 | TC1 | TC2 | CC1 | Additional Notes

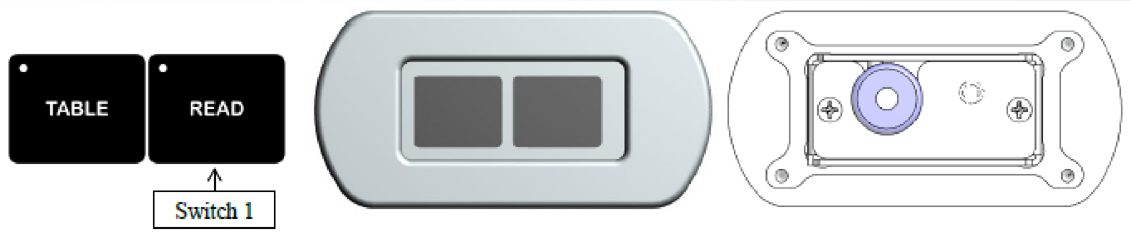
Cabin Management Checkout and Troubleshooting

Create a thorough test plan:

- Every switch and function is accounted for in your checklist.
- Include all CMS functions in your checklist (Lighting, Galley, Water, Audio, Video, Call, PA etc.)
- Test discrete switches as well as all functions on touchscreens/app.

MAIN MENU	SELECTION / SUB MENU	FUNCTION CONTROL
	TABLE	ON/OFF ____(√)
	READ	ON/OFF ____(√)

Pass	Fail



Cabin Management Checkout and Troubleshooting

Isolating the problem

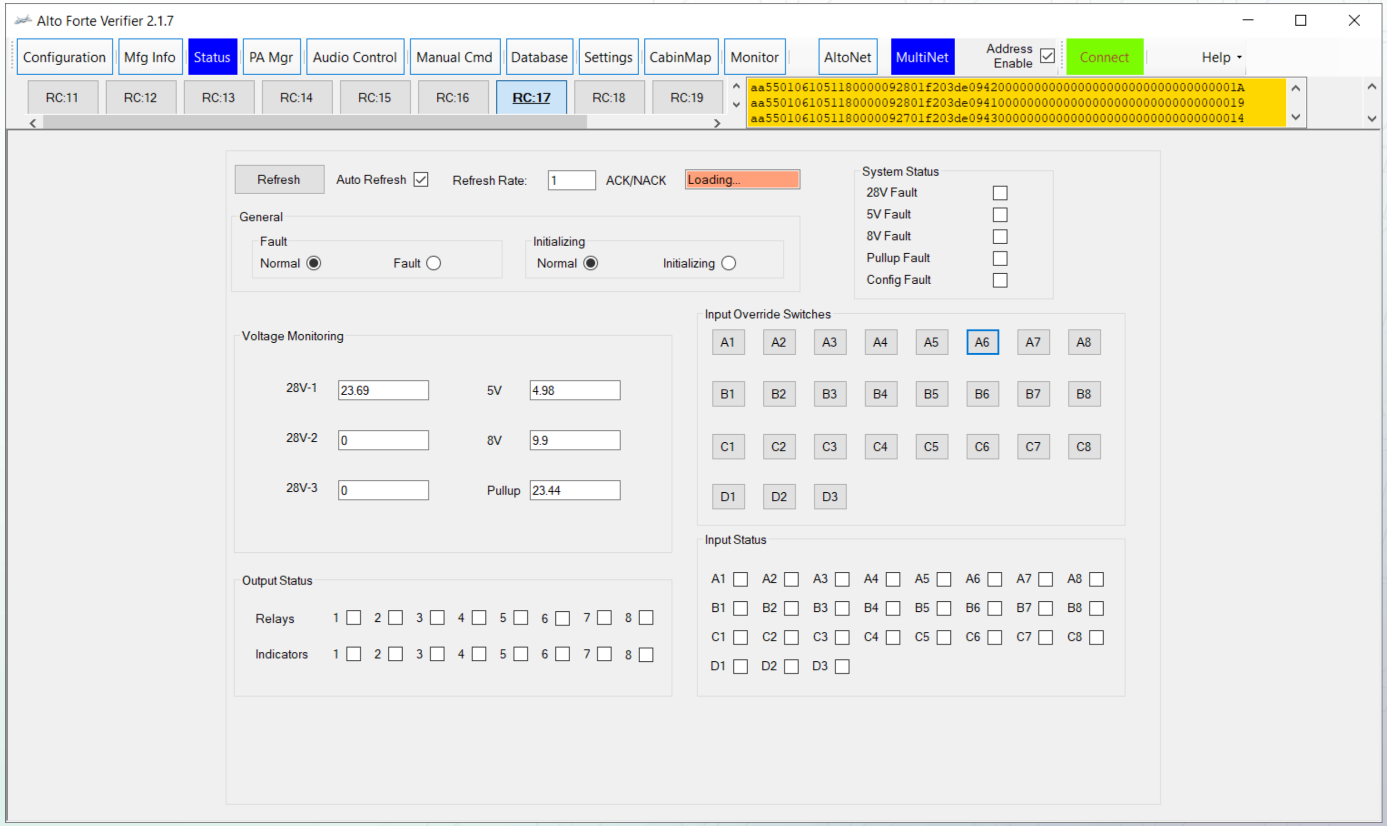
- If a switch doesn't work, in a discrete system, typically you can isolate the switch from the input to CMS and test both separately. ALTO switches output a discrete ground when pressed, so you can validate keypad on your bench with a meter independent from the aircraft.

- If a switch doesn't work in a CMS with databus backbone, typically diagnostic tools can tell you if issue lies on input or output side of the controller.

Cabin Management Checkout and Troubleshooting

Using Diagnostic Tools

- ALTO & Innovative Advantage Provide:
  - ALTO Forte Verifier software
  - IA AVDS Client software
  - ALTO Serial Maintenance Cables
  - Remote Troubleshooting Support
- ALTO software tutorials are available on YouTube [HERE](#).
- Innovative Advantage Academy



Cabin Management - Network Troubleshooting

Ethernet - widely used interface on aircraft to connect devices on a shared network, allowing them to send and receive data to one another

Common Uses:

- Cabin management command and control
- Internet connectivity systems
- Audio/video systems (IFE, cameras)
- Sensor and system data

Benefits:

- High speed / high bandwidth
- Scalable architecture
- Lower implementation cost
- Standardized communication
- Reduced wiring complexity

Ethernet Cabling

Types:

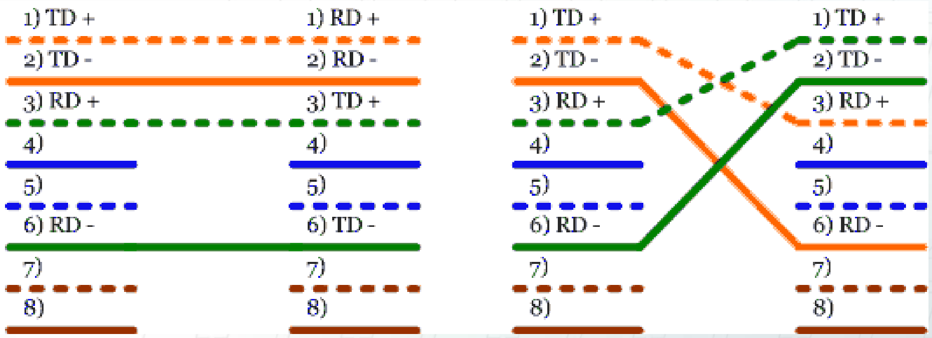
- Copper, fiber optic, Single Pair Ethernet
- Copper: Cat5, Cat6, Cat7, etc.
  - Shielded vs. unshielded cables
    - Aerospace is 100% shielded

Speeds:

- 10Mbps, 100Mbps, and 1000Mbps (1Gbps) most common
- 2.5Gbps, 10Gbps, and even higher now used

Pinouts:

- T568A and T568B standards, 100Base-T, and others
- Straight thru or crossover pinouts



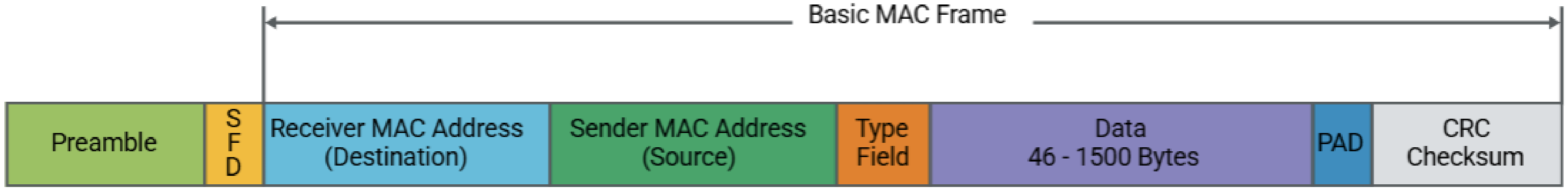
100Mbps Cable: Straight thru vs. crossover

1Gbps Cable: Straight thru vs. crossover

Ethernet and How it works

Ethernet Data is transmitted in frames that include:

- Destination address
- Source address
- Data payload
- Error checking



Ethernet Switches manage communication between devices

- Learn device addresses
- Direct traffic efficiently
- Reduce unnecessary network traffic

MAC Addresses

A **MAC address** is a **unique identifier** assigned to every device (network interface) for communication on a physical network segment

- You can think of it like the street address of a house — it is permanently associated with the hardware and typically does not change

MAC addresses are 48-bit identifiers

- Usually written as six pairs of hexadecimal digits separated by colons or hyphens
    - Example: 00:1A:2B:3C:4D:5E

Vendor Identification

- The first three bytes (24 bits) of a MAC address are called the Organizationally Unique Identifier (OUI)
- These prefixes are assigned to manufacturers by the Institute of Electrical and Electronics Engineers (IEEE)
    - Example: A MAC address beginning with C4:48:38 indicates a device manufactured by Satcom Direct

Broadcast Address

- The address 255.255.255.255 is a limited broadcast address.
- Packets sent to this address are delivered to every device on the local network but are not forwarded by routers

IP Addresses

An **IP address** is a numerical label **assigned to each device on a network**

It allows devices to identify each other and send data across networks

- Think of it like a mailing address for a device — it tells the network where data should be delivered.

A subnet mask divides an IP address into two parts:

- Network portion – identifies the network
- Host portion – identifies the device

The **subnet mask** determines **which devices** can communicate directly

- Devices within the same subnet communicate directly
- Devices outside the subnet communicate through a router

IP: 192.168.1.5
Subnet: /24 (255.255.255.0)
Network: 192.168.1.0
Host: 5

IP: 172.16.1.1
Subnet: /16 (255.255.0.0)
Network: 172.16.0.0
Host: 1

Capturing Network Traffic - Hardware

Networks normally send packets only to their intended destination, so most traffic on the network is not visible to your laptop.

To capture all packets on a network segment, you need one of the following:

- A network TAP
- A hub (not a switch)
- A managed switch with port mirroring (e.g., Netgear GS105Ev2)

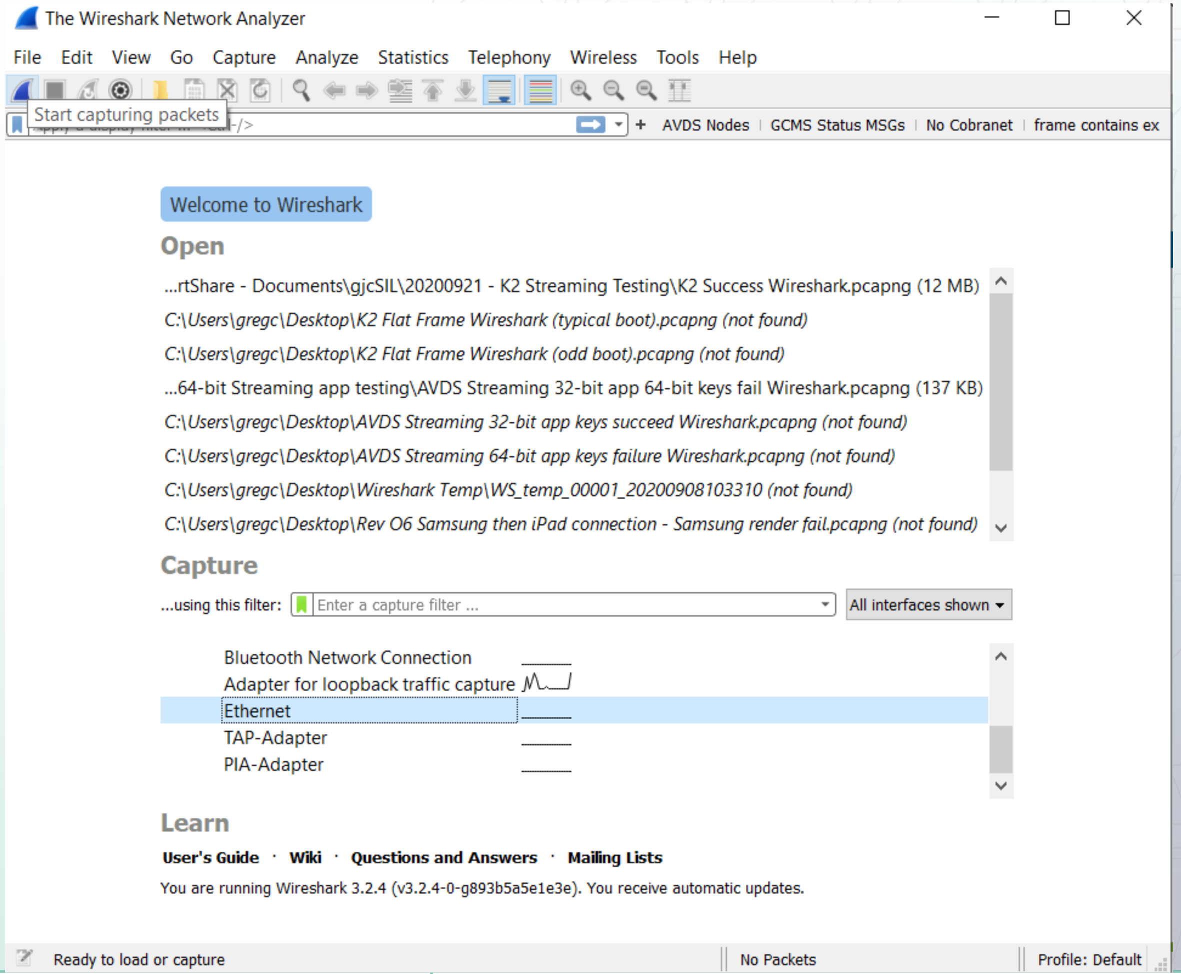
Notes:

- With a TAP, your monitoring port cannot transmit—your laptop is invisible to the network.
- With port mirroring, you may occasionally see duplicate packets.

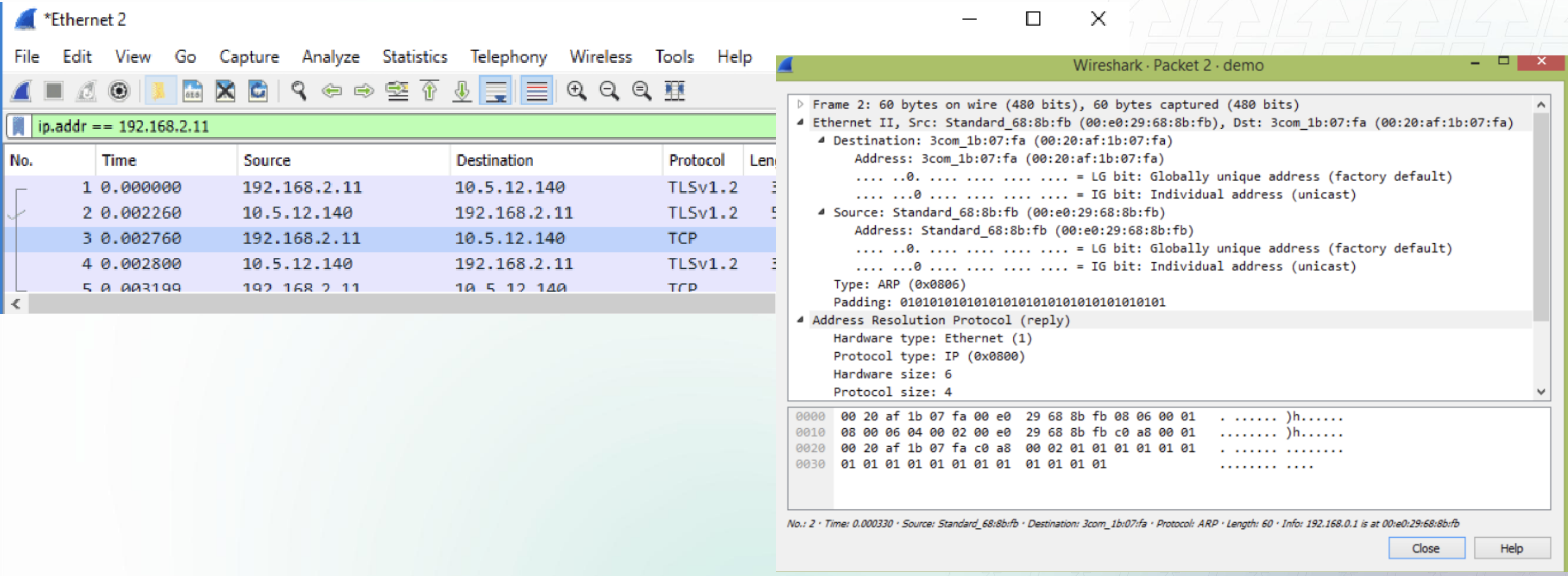
Placement matters:
You must monitor the specific network link carrying the traffic you want to capture.

Capturing Network Traffic - Software

- Wireshark is a software used to capture all network traffic from a specific port
- Most basic use:
  - Open Wireshark
  - Click the proper adapter to use
  - Click the blue fin to start a capture
  - Click stop when done
  - Save capture and email
- Lots more information: https://www.wireshark.org/learn



Wireshark Basics



Contacting Support Resources

ALTO	Innovative Advantage
+1-972-980-4890	+1-888-838-8230
support@AltoAviation.com	support@In-Advantage.com

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